

Output 1 - Semantic Representation in Customer Support Chatbots

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Query ID: Q1
Semantic Representation: ACTIVATE(Roaming, Customer)
Action: ACTIVATE
Object: Roaming
Actual Intent: Activate Roaming
Predicted Intent: Activate Roaming
Status: Correct

Query ID: Q2
Semantic Representation: DEACTIVATE(CallerTune, Customer)
Action: DEACTIVATE
Object: CallerTune
Actual Intent: Deactivate Caller Tune
Predicted Intent: Activate Caller Tune
Status: ERROR

Query ID: Q3
Semantic Representation: QUERY(DataBalance, Customer)
Action: QUERY
Object: DataBalance
Actual Intent: Query Data Balance
Predicted Intent: Query Data Balance
Status: Correct

Query ID: Q4
Semantic Representation: ACTIVATE(5GService, Customer)
Action: ACTIVATE
Object: 5GService
Actual Intent: Activate 5G Service
Predicted Intent: Activate 5G Service
Status: Correct

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Correct Predictions: 3
Total Queries: 4
Chatbot Accuracy: 75.0 %
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Output 2 - First-Order Predicate Calculus for Smart Manufacturing

FIRST-ORDER PREDICATE CALCULUS

Production Data:

Active(M1)

Active(M2)

Maintenance(M3)

Active(M4)

Production Relationships:

Produces(M1, Gear)

Produces(M2, Bolt)

Produces(M3, Gear)

Produces(M4, Shaft)

Predicate Inference:

Active(M1) $\rightarrow$ Producing(M1)

Produces(M1,Gear) AND Active(M1) $\rightarrow$ Available(Gear)

Active(M2) $\rightarrow$ Producing(M2)

Produces(M2,Bolt) AND Active(M2) $\rightarrow$ Available(Bolt)

Maintenance(M3) $\rightarrow$ NOT Producing(M3)

Active(M4) $\rightarrow$ Producing(M4)

Produces(M4,Shaft) AND Active(M4) $\rightarrow$ Available(Shaft)

Currently Available Products:

Bolt

Gear

Shaft

Products Affected by Maintenance:

Gear

Conclusion:

Gear production is affected because M3 is under maintenance.

Output – 3 Word Sense Disambiguation

Query: Apple accessories

Ambiguous Word: Apple

Possible Senses: Fruit, Technology Brand

Clicked Result: iPhone Charger

Predicted Sense: Technology Brand

Expected Sense: Technology Brand

Status: Correct

Query: Mouse wireless

Ambiguous Word: Mouse

Possible Senses: Animal, Computer Device

Clicked Result: Bluetooth Mouse

Predicted Sense: Computer Device

Expected Sense: Computer Device

Status: Correct

Query: Java tutorial

Ambiguous Word: Java

Possible Senses: Island, Programming Language

Clicked Result: Coding Lessons

Predicted Sense: Programming Language

Expected Sense: Programming Language

Status: Correct

Query: Python course

Ambiguous Word: Python

Possible Senses: Snake, Programming Language

Clicked Result: Software Development Training

Predicted Sense: Programming Language

Expected Sense: Programming Language

Status: Correct

Correct Sense Predictions: 4

Total Queries: 4

WSD Accuracy: 100.0 %

Output 4 - Syntax-Driven Semantic Analysis in Healthcare

SYNTAX-DRIVEN SEMANTIC ANALYSIS

Sentence: Doctor prescribed medicine to patient.

Syntactic Structure: Subject - Verb - Object

Subject: Doctor

Verb: prescribed

Object: medicine

Semantic Roles:

Doctor -> Agent

Medicine -> Theme

Patient -> Recipient

Sentence: Patient reported severe headache.

Syntactic Structure: Subject - Verb - Object

Subject: Patient

Verb: reported

Object: headache

Semantic Roles:

Patient -> Experiencer

Headache -> Symptom

Sentence: Nurse monitored patient continuously.

Syntactic Structure: Subject - Verb - Object

Subject: Nurse

Verb: monitored

Object: patient

Semantic Roles:

Nurse -> Agent

Patient -> Patient/Target

Sentence: Medicine reduced blood pressure.
Syntactic Structure: Subject - Verb - Object
Subject: Medicine
Verb: reduced
Object: blood pressure
Semantic Roles:
Medicine -> Cause/Agent
Blood Pressure -> Theme

ROLE VALIDATION

Doctor -> Agent
Medicine -> Theme/Cause
Patient -> Recipient/Experiencer/Target
Headache -> Symptom
Nurse -> Agent
Blood Pressure -> Theme

Potential Parsing Errors:

1. Incorrect subject identification can produce incorrect Agent roles.
2. Incorrect object identification can change the Theme or Patient role.
3. Prepositional phrase errors may confuse Recipient and Instrument roles.
4. Medical terminology may cause incorrect semantic interpretation.

Recommended Improvements:

1. Use dependency parsing.
2. Use medical NLP models.
3. Use semantic role labeling.
4. Use medical dictionaries and domain-specific ontologies.
5. Validate results using clinical context.